

Problem Statement 1:

Our coffee sales and profits have been fluctuating over the past two years—can you help us understand which coffee types or roast types are driving or hurting our performance so we can improve overall profitability?

Questions:

1. Which coffee types are driving the highest profit and profit margins?
2. Which coffee types are underperforming and reducing overall profitability?
3. How does each roast type contribute to sales and profit performance?
4. Which roast type is most effective in maximizing profitability?
5. What combination of coffee type and roast type yields the best returns?
6. Why did sales and profit decline significantly from 2021 to 2022?
7. Is the performance issue driven by reduced demand, product mix, or pricing?
8. Which products should be prioritized to improve overall profitability?
9. Which products should be optimized, repositioned, or discontinued?
10. How can we adjust the product mix to stabilize and grow revenue and profit?

Dashboard:

1. **KPI Cards (Total Sales, Profit, Quantity, Margin)** → What is the overall business performance at a glance?
2. **Sales & Profit Margin % by Year (Line Chart)** → How have sales and profitability trends changed over time?
3. **Total Sales/Profit by Coffee Type (Bar Chart)** → Which coffee types are driving revenue and profit?
4. **Profit Margin % by Coffee Type (Bar Chart)** → Which coffee types are most and least profitable?
5. **Total Sales/Profit by Roast Type (Bar Chart)** → Which roast types contribute most to performance?
6. **Sales vs Profit vs Quantity by Coffee Type (Scatter Plot)** → How do coffee types compare across volume, revenue, and profitability?
7. **Top Products Table** → Which individual products contribute most to sales and margin?
8. **Monthly Performance Table (Year Reports)** → How does performance vary month-to-month and where are fluctuations occurring?
9. **Coffee Type × Roast Type Stacked Bar (Year Reports)** → Which combinations of coffee type and roast type drive results?
10. **Parameter Toggle (Sales/Profit/Quantity)** → How does performance change when switching between key metrics?

 Insights

- Liberica is the highest profit and margin contributor.
- Excelsa drives the highest sales volume with solid profitability.
- Arabica delivers average performance with lower margins.
- Robusta is the weakest performer across sales, profit, and margin.
- Light Roast generates the highest sales and profit.
- Dark Roast contributes the least to overall performance.
- Profit margin is stable (~10%) despite revenue fluctuations.
- Significant decline in sales and profit observed in 2022.
- Performance drop is driven by reduced volume, not margin erosion.

 Recommendations

- Focus on scaling Liberica to maximize profitability.
- Leverage Excelsa to maintain and grow volume.
- Optimize or reposition Arabica to improve margins.
- Reduce or phase out Robusta due to poor returns.
- Prioritize Light Roast in production and marketing.
- Reassess demand and strategy for Dark Roast.
- Improve product mix toward high-margin combinations.
- Investigate causes of 2022 decline and address demand gaps.
- Launch targeted promotions on high-performing products.
- Optimize costs in low-margin segments to improve overall profitability.